

Preliminary Data Analyses

Load in libraries

```
library(tidyverse)
library(plyr)
library(stargazer)
library(pheatmap)
library(corrplot)
```

Read in cleaned dataset

```
data <- read_csv("data/data_clean.csv") |>
  data.frame()
```

Creating descriptive statistics table for phthalate values

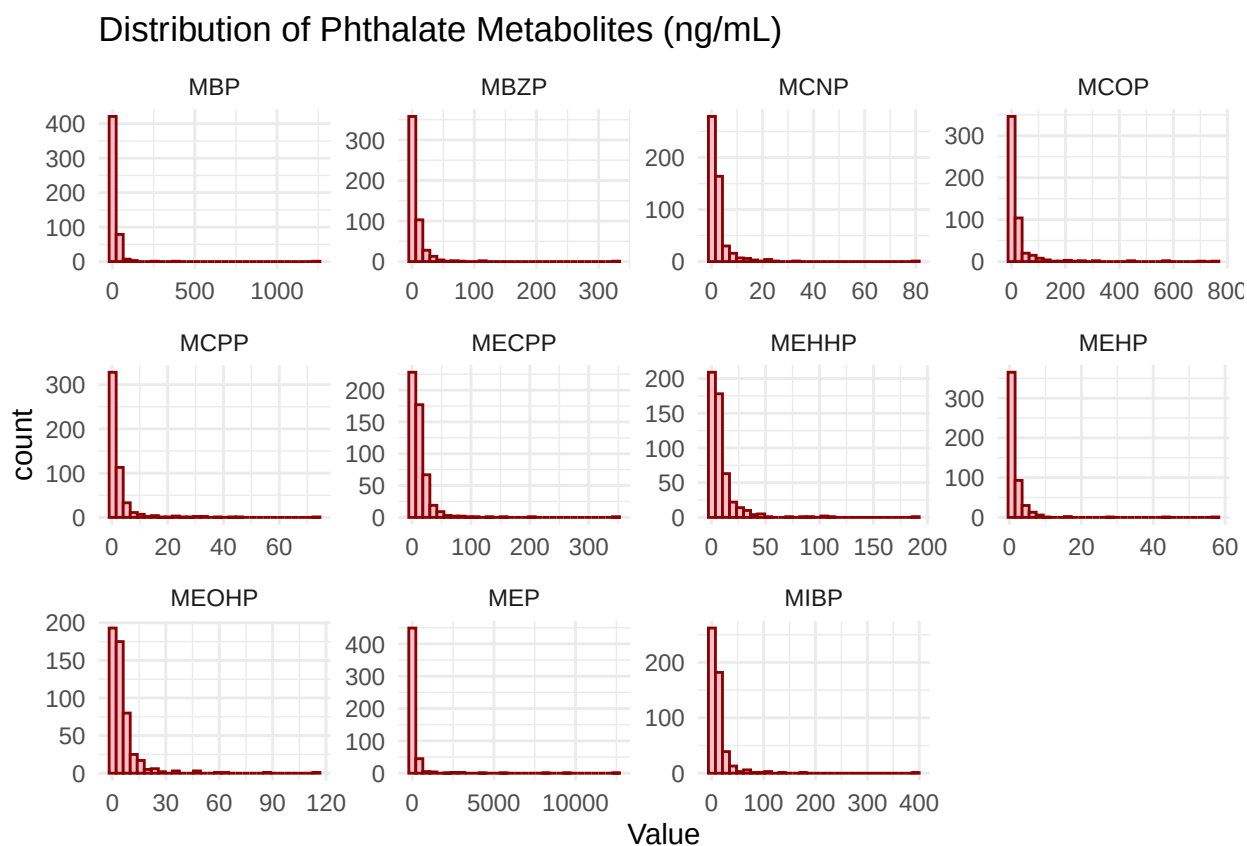
```
stargazer(data |>
  select(MCNP:MBZP),
  type = "text",
  summary = TRUE,
  summary.stat = c("n", "mean", "p25", "median", "p75"))
```

```
##
## =====
## Statistic  N   Mean   Pctl(25) Median Pctl(75)
## -----
## MCNP       513  2.713    0.800    1.400    2.800
## MCOP       513 26.221    3.000    6.900   18.600
## MECP       513 13.201    3.600    7.200   15.400
## MBP        513 17.543    4.100    9.000   16.900
## MCPP       513  2.688    0.500    1.000    2.200
## MEP        513 192.551   13.700   34.700   91.200
## MEHHP      513  9.013    2.200    4.700    9.900
## MEHP       513  1.768    0.570    0.570    1.700
## MIBP       513 12.581    3.400    7.200   14.000
## MEOHP      513  5.634    1.400    3.000    6.500
## MBZP       513  7.599    1.200    2.900    7.900
## -----
```

Histograms to show the distribution of values for the phthalate metabolites

```
data_long <- data %>%  
  pivot_longer(cols = MCNP:MBZP,  
               names_to = "Metabolite",  
               values_to = "Value")  
  
ggplot(data_long, aes(x = Value)) +  
  geom_histogram(fill = "pink", color = "darkred") +  
  facet_wrap(~ Metabolite, scales = "free") +  
  labs(title = "Distribution of Phthalate Metabolites (ng/mL)") +  
  theme_minimal()
```

'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.

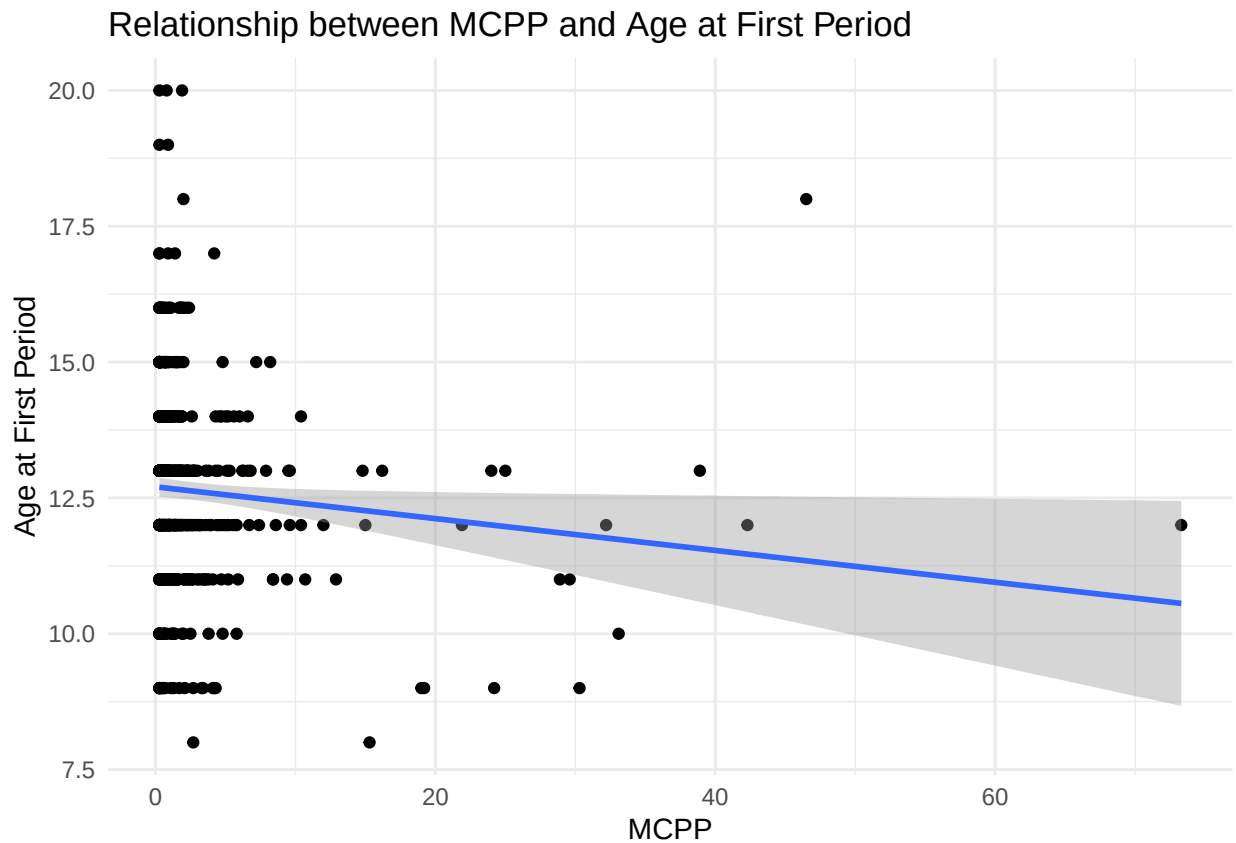


Plot the relationship between MCP and menstruation

```
ggplot(data,  
       aes(x = MCP, y = Age_FirstPeriod)) +
```

```
labs(title= "Relationship between MCPP and Age at First Period",
      x = "MCPP",
      y = "Age at First Period") +
geom_point() +
geom_smooth(method = "lm", se = TRUE) +
theme_minimal()
```

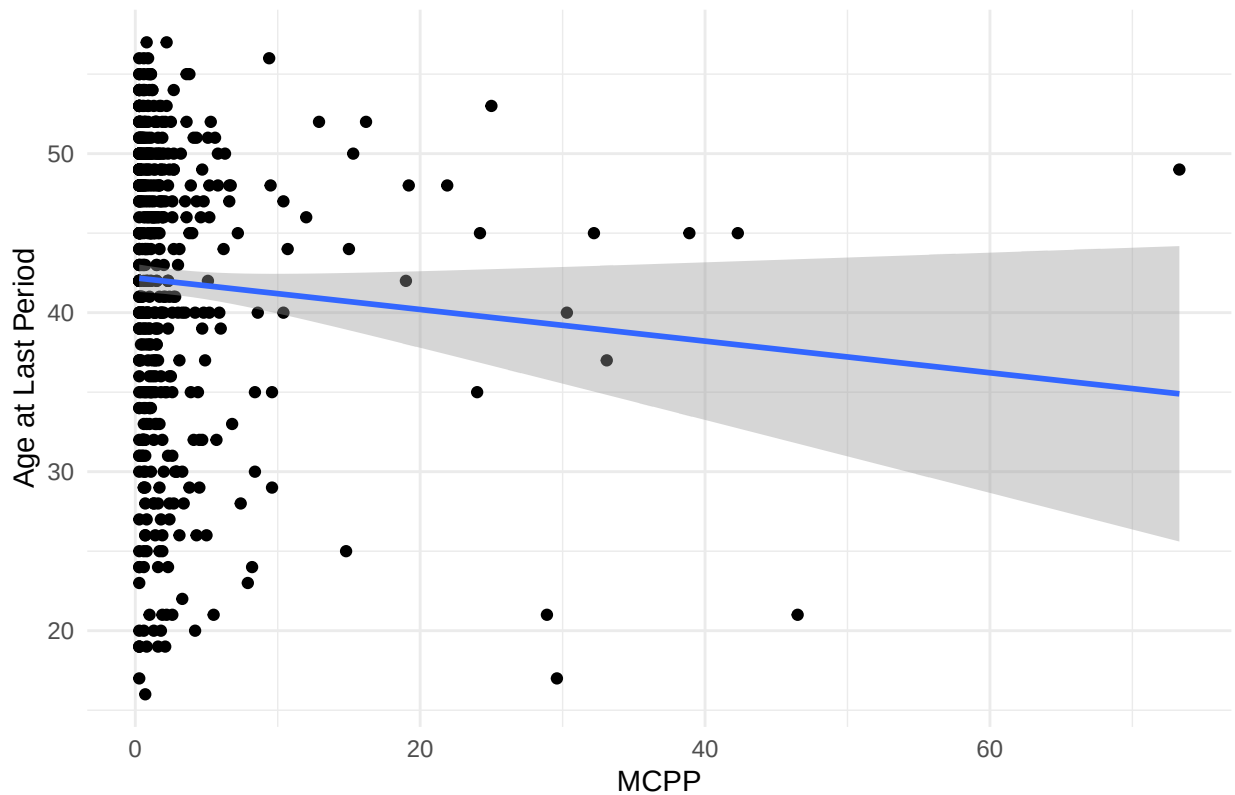
'geom_smooth()' using formula = 'y ~ x'



```
ggplot(data,
      aes(x = MCPP, y = Age_LastPeriod)) +
labs(title= "Relationship between MCPP and Age at Last Period",
      x = "MCPP",
      y = "Age at Last Period") +
geom_point() +
geom_smooth(method = "lm", se = TRUE) +
theme_minimal()
```

'geom_smooth()' using formula = 'y ~ x'

Relationship between MCPP and Age at Last Period



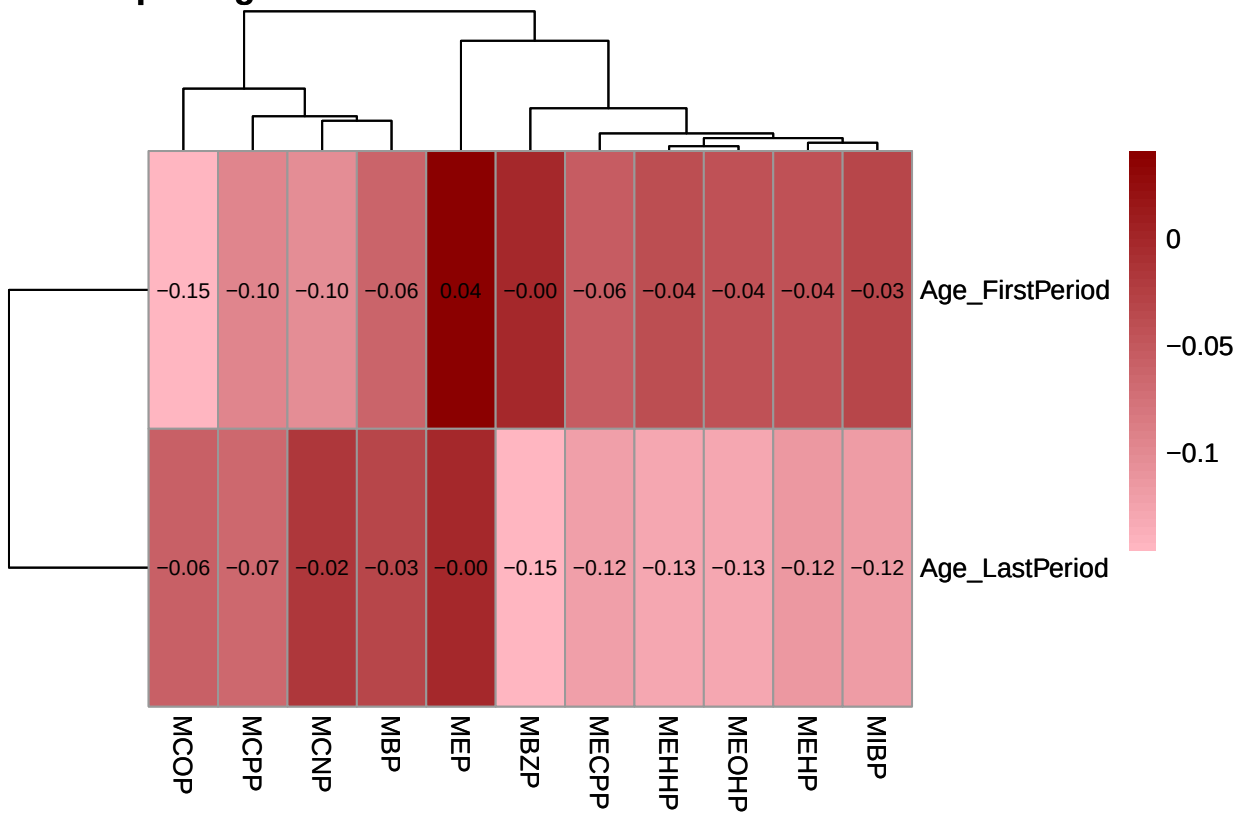
Heatmap of correlations

```
correlation_matrix = cor(data |>
  select(Age_FirstPeriod, Age_LastPeriod),
  data |>
  select(MCNP:MBZP)
)

corr_heatmap = pheatmap(corr_matrix,
  color = colorRampPalette(c("lightpink", "darkred"))(50),
  number_color = "black",
  display_numbers = TRUE,
  cluster_rows = TRUE,
  cluster_cols = TRUE,
  main = "Heatmap of Age at First or Last Period x Phthalate Correlations")

corr_heatmap
```

Heatmap of Age at First or Last Period x Phthalate Correlations



Correlation matrix

```
corrplot(cor(data |> select(MCNP:MBZP)), method = "circle", type = "full")
```

